Leakage-Free Cervical Cell Classification Using Dynamic Cross-Window Attention and Quantitative Explainability

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Abstract—Reliable Cervical cancer remains a major health concern among women worldwide, particularly in regions where access to regular and expert screening is limited. Pap smear cytology is an established screening method, but accurate cell interpretation depends heavily on trained cytologists and can be affected by observer variability and workload. Automated cervical cell classification using deep learning can support cytologists as a second reader. In this work, we present a Swin Transformer classifier, trained from scratch, with two new components and one new evaluation. *Dynamic Cross-Window Attention* (DCWA) makes window connectivity content-dependent. It builds an input-dependent affinity between windows and adds the gathered context to every token through a learned gate. *Adaptive Dynamic Focal Loss* (ADFL) adapts to class difficulty during training. It sets class weights and per-class focusing from a running difficulty estimate, and it raises its global focusing parameter over the epochs. A *quantitative explainability evaluation* tests Grad-CAM against a random-order baseline. It uses deletion and insertion curves and a paired Wilcoxon test. We evaluated on SIPaKMeD (five classes, 5,015 images) with a frozen, cluster-wise split that keeps all images of a cluster in one partition. On 734 held-out images, the model trained for up to 100 epochs reaches 90.87% accuracy, 91.24% macro-F1, 91.21% sensitivity and 97.69% specificity. Four baselines trained from scratch for 16 epochs reach at most 88.28% accuracy and 88.71% macro-F1. In a three-seed, 16-epoch ablation, DCWA adds 0.41 points of accuracy, which lies within seed variability. ADFL lowers accuracy by 1.50 points under this short schedule. Grad-CAM beats random ordering on deletion (AUC 0.466 vs. 0.546, $p < 10^{-15}$). Overall, the study evaluates the proposed classifier not only through classification performance but also through controlled ablation and quantitative explainability analysis, providing evidence for its potential use as a second-reader system to support cytological assessment.

Key Words —Cervical cancer, Pap smear, SIPaKMeD, Swin Transformer, cross-window attention, focal loss, Grad-CAM, explainable AI, deletion and insertion curves, clinical decision support.

I. Introduction

Cervical cancer remains one of the most common cancers in women worldwide [1]. The burden falls hardest because screening capacity is limited. Pap smear screening works, but it relies on trained cytotechnologists who must examine thousands of cells per slide manually. The work is repetitive, and observers often disagree. Automated cell classification can support doctors as a second reader, help prioritise slides for review and make screening more accessible in areas where specialists are limited [3].

Deep learning now dominates this task. Hierarchical vision transformers such as Swin [6] suit it well. The windowed self-attention of Swin captures local cell morphology at a manageable cost. The publicly available SIPaKMeD dataset [2] is commonly used for this evaluation model. It has five classes: Dyskeratotic, Koilocytotic, Metaplastic, Parabasal and Superficial-Intermediate. However, there are still some important limitations. Existing studies do not always provide a clear picture of how well these models would perform in real clinical settings.

Current cervical smear analysis pipelines suffer from four critical limitations of their clinical deployment. First, random image-level splitting introduces data leakage and artificially inflates performance because crops from the same microscope field act as near-duplicates. Second, fixed window attention restricts token interactions locally, this fails to capture essential global context such as cell cluster layouts and staining patterns across the smear. Third, uniform loss functions ignore varying class difficulties, struggling with the severe visual and morphological overlap characteristic of koilocytotic and metaplastic cells. Fourth, existing model explanations rely on a few qualitative heatmaps without exact comparison against established interpretability baselines. as a few heat maps. Authors seldom test them against a baseline. It is unclear whether they show what the model uses.

This paper makes four contributions.

(1) **Dynamic Cross-Window Attention (DCWA): content-dependent window connectivity.** In Swin [6], CSWin [7] and Twins [8], the architecture fixes which windows exchange information. DCWA lets each image decide. It measures how window modifiers relate to each other through a multi-head affinity score. It gathers cross-window context and adds it to each token through a sigmoid gate. A zero-initialised output projection makes the network start as plain Swin. We insert DCWA into every block of the first three stages.

(2) **Adaptive Dynamic Focal Loss (ADFL): class difficulty drives the loss.** Focal loss [11] uses one fixed focusing exponent and fixed class weights. Class-balanced loss [12] corrects for class frequency, not for difficulty. ADFL measures the difficulty of each class as training proceeds. It uses that estimatimation to set the weight and the focusing exponent of every class. A global focusing parameter also grows over the epochs.

(3) **Quantitative explainability evaluation.** The model score Grad-CAM with deletion and insertion curves and compare each score with a random pixel ordering on every test image. To prove it , we run a paired Wilcoxon test. If the explanation doesn't beat random chance, we don't trust it.

(4) **A leakage-free, reproducible protocol.** The model use a deterministic cluster-wise split frozen in a manifest with a SHA-256 hash per image. It check for group leakage and

duplicates. Augment the training set only. It packages the best model with its test images and a verification script.

The main results are as follows. With a long schedule, the full model reaches 90.87% test accuracy and 91.24% macro-F1. It represents the top performance achieved across all experimental runs in this study, the integration of the two proposed components does not yield performance gains over the standard Swim Transformer baseline. DCWA gives a marginal accuracy improvement that is statistically inconclusive. While ADFL slightly degrades classification performance. Grad-CAM proves highly informative during the deletion test, lending validity to this transparent evaluation as a core contribution of our work. It clarifies what has already been proven evidence from what needs to be shown. We also describe how cytologists could use the system (Section VII) and how explanations can build trust (Section VII-B).

Section II reviews related work. Section III describes the data and protocol. Section IV presents the method. Section V gives the experimental setup and Section VI the results. Section VII covers clinical relevance. Section VIII discusses the findings and limitations. Section IX concludes.

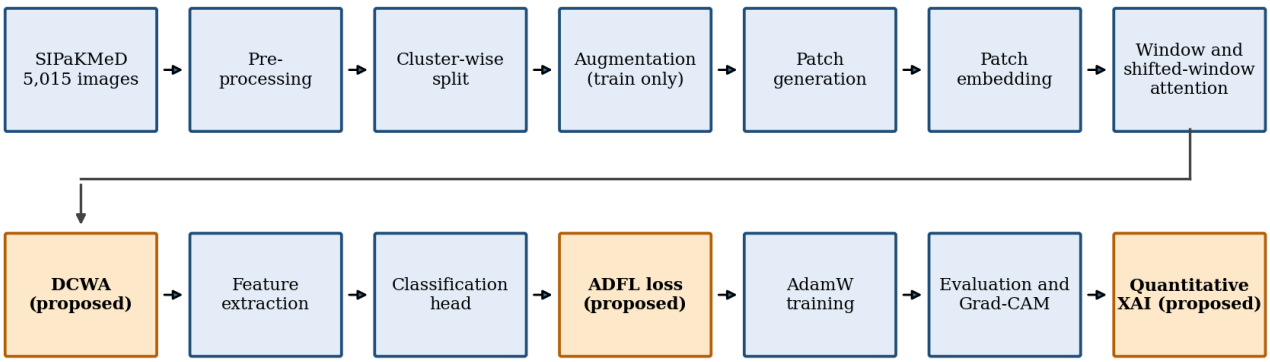


Fig. 1. Workflow of the proposed system. Orange blocks are the proposed components: Dynamic Cross-Window Attention (DCWA), Adaptive Dynamic Focal Loss (ADFL) and the quantitative explainability evaluation.

II. Related Work

A. Automated cervical cytology

Early screening systems used hand-crafted nuclear and cytoplasmic features. These included size, shape, texture and chromatin statistics. Classical classifiers then labelled the cells. The Survey [3] covers these methods. Convolutional networks replaced hand-crafted features. With transfer learning from natural images, they reached high accuracy on small cytology datasets. SIPaKMeD [2] provides isolated cell crops and cluster images in five categories. It is now among the most used datasets.

Published five-class accuracies are often in the high 90% range. Authors obtain them with ImageNet-pretrained backbones, ensembles or multi-branch designs. These figures are hard to compare. Studies differ in split policy (image-level or group-level), pretraining, model count and the order of augmentation and splitting. Many do not state these choices. This model compares only against baselines trained under one identical protocol.

Evaluation protocols. In medical imaging, splitting by image is a known source of optimistic bias when several images come from one patient, slide or field. The unit of splitting should be the group that shares acquisition. In SIPaKMeD, file names carry a shared prefix for crops of the same field. This makes group-level splitting possible, and our protocol uses it. We cannot quantify the optimism this removes, because we ran no image-level split for comparison.

B. Vision transformers and window connectivity

ViT [5] adapted the transformer [4] to images. It splits an image into patches and applies global self-attention. Global attention scales quadratically with token count, and ViT needs large amounts of data. Swin [6] restricts attention to local windows. It alternates regular and shifted partitions so that information crosses window borders. It builds a multi-scale hierarchy through patch merging. Later designs change the window shape or mix in global attention. CSWin [7] uses cross-shaped windows. Twins [8] alternates local and sub-sampled global attention. In all of them, the architecture fixes the connectivity pattern. It does not depend on the image. DCWA differs on exactly this point. *The content of the current image decides which windows exchange information.* The decision comes from a learned affinity between window descriptors. The cost grows with the square of the number of windows, not of tokens.

C. Losses for hard and imbalanced classes

Focal loss [11] multiplies cross-entropy by $(1 - p_t)^{\gamma}$. Well-classified samples then contribute less. It uses one fixed γ and fixed class weights. Class-balanced loss [12] re-weights classes by their effective number of samples. This addresses frequency but not difficulty. ADFL combines both signals and uses a frequency prior and a running estimate of per-class difficulty. The difficulty estimate modulates both the class weight and the focusing exponent. A curriculum raises the global focusing parameter over training.

D. Evaluating explanations

Grad-CAM [13] is the most common visual explanation for classification networks. However, it does not prove

faithfulness. Saliency maps can look convincing while ignoring the model [15]. Perturbation metrics measure how the predicted probability changes when pixels are removed (deletion) or revealed (insertion) in the order an explanation gives. RISE [14] popularised them. These metrics have known pitfalls. Masked images lie outside the training distribution [16], so an absolute AUC is hard to interpret. A comparison with random ordering on the same images gives the metric a reference point. We use that comparison.

III. Dataset and Evaluation Protocol

A. SIPaKMeD dataset

SIPaKMeD [2] contains microscope images of Pap smears. Cytopathologists labelled them into five categories: Dyskeratotic, Koilocytotic, Metaplastic, Parabasal and Superficial-Intermediate. The first three are abnormal or benign-abnormal changes. The last two are normal. The model uses all 5,015 images, both isolated-cell crops and cluster images. It takes each class name from the folder path and check that it is unambiguous. The held-out test set has 734 images (Table I).

TABLE I
Class distribution of the held-out test set

Class	Images	Share (%)
Dyskeratotic	152	20.7
Koilocytotic	159	21.7
Metaplastic	153	20.8
Parabasal	129	17.6
Superficial-Intermediate	141	19.2
Total	**734**	**100.0**

B. Deterministic cluster-wise split

Image file names share a numeric prefix when they come from the same microscope field or cluster (for example 001_01 and 001_02). It defines a *group* as the pair (class, prefix). It assigns each whole group to exactly one partition. Within each class, it orders the groups by a random permutation with a fixed seed (42). It cut the images order into about 70% training, 15% validation and 15% test images.

It writes the result once to a manifest. This stores the path, class, group, split and SHA-256 hash of every image. Later runs load the manifest instead of recomputing the split. The partition therefore cannot change between experiments. The code asserts that no group appears in more than one split. It also counts byte-identical images across splits. This protocol is stricter than an image-level split. The model expects it to give lower but more realistic accuracy.

C. Pre-processing and augmentation

The model resizes all images to 224 × 224 pixels with area interpolation and normalises them with the ImageNet channel means and standard deviations. The augment *after* the split and *only the training set*. The augmentations are horizontal and vertical flips (p = 0.5 each), random 90° rotation (p = 0.5), an affine transform with scale in [0.9, 1.1] and rotation within ±15° (p = 0.5), and mild colour jitter (brightness, contrast and saturation 0.1, hue 0.02; p = 0.3). Applying augmentation before the dataset split could place transformed versions of the same image into different partitions, leading to data leakage. Validation and test images receive only resizing and normalisation.

D. Patch generation

Swin cuts an image into non-overlapping 4 × 4 patches. A 224 × 224 image yields 56 × 56 = 3,136 patches. The 5,015 images yield about 15.7 million patches. The patch-embedding layer creates them on the fly, so we store nothing. We verified with a patchify/unpatchify pair that the operation is exactly reversible on a validation batch and also verified that the token grid of every stage (56, 28, 14 and 7) fits the 7 × 7 window. The four stages contain 64, 16, 4 and 1 windows.

E. Reproducibility package

After training, we package the best checkpoint with the test images, a test manifest, the source code and an index file. The index links the model hash, the test-manifest hash and the split-manifest hash. A stand-alone script rechecks all hashes and reproduces the test metrics without any training data.

IV. Proposed Method

A. Overview

Fig. 1 shows the workflow. The model splits an image into patches and embeds them. Four Swin stages with depths 2, 2, 6 and 2 process the tokens. Each block applies window self-attention, alternating with a shifted-window version. In the first three stages, a DCWA module follows each block. The model pools the features and classifies them with a linear head. ADFL trains the network. After training, we evaluate the model on the test set. We explain its predictions with Grad-CAM and score the explanations with the proposed procedure.

B. Swin backbone

The backbone follows Swin-T. The embedding dimension is 96. The attention heads are (3, 6, 12, 24), the window size is 7 and the MLP ratio is 4. Stochastic depth rises linearly to 0.1. Patch merging between stages gives channel widths of 96, 192, 384 and 768. Window attention uses a learned relative position bias B:

$$Attn(Q,K,V) = softmax(QK^T/\sqrt{d} + B)V \quad (1)$$

In the last stage, the 7 × 7 grid fits in one window, so we disable the shift. The final representation is the mean of the 7 × 7 tokens after layer normalisation.

C. Dynamic Cross-Window Attention (DCWA)

Novelty: content-dependent window connectivity. Let $y \in \mathrm{R}^{nW \times N \times C}$ be the window-attention output of one block for one image. Here nW is the number of windows and N = 49 is the number of tokens per window. DCWA (Fig. 2) acts on y before the windows merge back into the feature map. We first normalise: $h = \mathrm{LN}(y)$. We then summarise each window by the mean of its tokens:

$$d_w = (1/N)\, \Sigma_t\, h_{w,t} \quad (2)$$

For each of the H heads, we compute queries, keys and values from the representations. The corelation between windows is an nW × nW matrix per head:

$$A^h = softmax(\, Q^h(K^h)^T / \sqrt{d_h}\,) \quad (3)$$

A depends on the descriptors, so the strength of each connection changes from image to image. Swin's fixed shift

pattern cannot do this. Window w receives the affinity-weighted sum of all value vectors, projected by W_o:

$$c_w = W_o \cdot Concat_h(\Sigma_j A^h_{w,j} v^h_j) \quad (4)$$

A token-wise sigmoid gate $g = \sigma(W_g h)$ then decides how much of this context each token and channel accepts:

$$\hat{y}_{w,t} = y_{w,t} + g_{w,t} \odot c_w \quad (5)$$

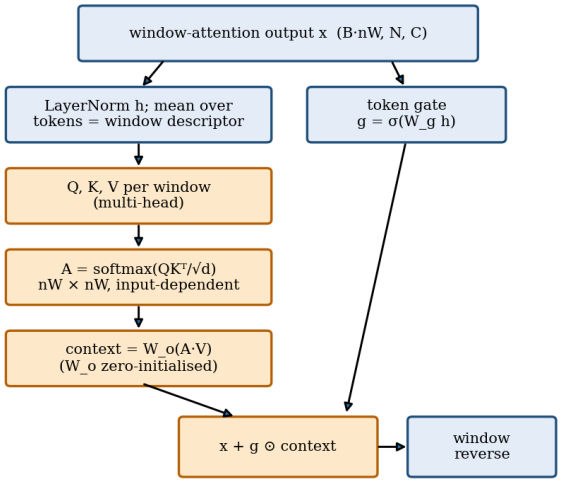


Fig. 2. Structure of the DCWA module. Orange blocks are the proposed operations. The window descriptors produce an input-dependent window-to-window affinity matrix. The resulting context is gated into every token.

W_o and its bias start at zero. At initialisation, $\hat{y} = y$, so the network equals plain Swin. DCWA departs from the baseline only if training finds it useful. We apply DCWA in every block of stages 1–3, ten blocks in total (2 + 2 + 6). Stage 4 has a single window, so there is nothing to connect. The affinities cost $O(nW^2C)$ per block. This is small next to token-level attention ($nW = 64$ in stage 1).

Cost. Each DCWA block has five C × C linear layers (queries, keys, values, output and gate) and a layer normalisation. That is $5C^2 + 7C$ parameters. Over the ten blocks (C = 96, 192 and 384 with 2, 2 and 6 blocks), DCWA adds about 4.9 million parameters. The corelation computation adds almost no compute. The token-wise gate dominates the extra cost. It is one C × C product per token, about one quarter of the four C × C projections in window attention.

Design rationale. These are design choices, not experimentally verified results. Mean pooling provides a permutation-invariant representation at a fixed computational cost. We derive descriptors from post-attention features, so the affinity captures information already identified by local attention. The softmax over windows makes the received context a convex combination, so windows compete for influence. The gate enables the network to reduce cross-window information for each token and channel. The zero-initialised projection guarantees that DCWA cannot harm the network at the start of training.

D. Adaptive Dynamic Focal Loss (ADFL)

Novelty: the loss adapts to class difficulty as training runs. Let p_i be the predicted probability of the true class of sample i. Let n_c be the number of training images of class c. Let B be the batch size. ADFL is

$$L = -(1/B) \Sigma_i \alpha_c (1 - p_i)^{\gamma_{-\{c}} \log p_i, \quad c = y_i \quad (6)$$

It has two adaptive parts. *Adaptive class weights.* We keep a running difficulty d_c for each class. It is an exponential moving average ($\beta = 0.9$, initial value 0.5) of the mean of $(1 - p_i)$ over the class-c samples in each batch. Classes absent from a batch are not updated. The relative difficulty is $r_c = d_c / \text{mean}(d)$. The weight combines a frequency prior $\pi_c \propto 1/\sqrt{n_c}$ (normalised to mean 1) with the difficulty. We clip the product to [0.5, 2.5] and renormalise to mean 1:

$$\alpha_c = norm(clip(\pi_c \cdot r_c, 0.5, 2.5)) \quad (7)$$

Dynamic focusing. The global focusing parameter follows a curriculum over epochs $e = 0, ..., T - 1$:

$$\gamma_e = \gamma_{min} + (\gamma_{max} - \gamma_{min}) \cdot e/(T - 1) \quad (8)$$

We set $\gamma_{min} = 0.5$ and $\gamma_{max} = 3$. Each class gets $\gamma_c = \gamma_e \cdot \text{clip}(r_c, 0.5, 2)$, so harder classes receive stronger focusing. At the beginning of training, the loss behaves like a weighted cross-entropy. Later, it focuses more on difficult samples and challenging classes. We compute this difficulty estimates without gradients. Thus, ADFL introduce no learnable parameters. Note that γ_e depends on the total number of epochs T. A small T compresses or shortens the curriculum. This is relevant to ablation study in Section VI.

Design rationale. The square-root inverse-frequency prior sits between no re-weighting and full inverse-frequency weighting. Full weighting tends to over-amplify rare classes. The moving average smooths the difficulty estimate, because a batch of 32 images holds only a few samples per class. Clipping stops any one class from dominating. As with DCWA, these are intentions. Section VI tests their effect.

Algorithm 1. Training and model selection

Input: frozen split manifest; model f_θ; ADFL; epochs T; patience
for e = 0, ..., T − 1 **do**
  set γ_e by Eq. (8)
  for each augmented training batch (x, y) **do**
    p = softmax(f_θ(x)); update d_c, r_c, α_c, γ_c (no gradient)
    L by Eq. (6); AdamW step with gradient clipping
  evaluate macro-F1 on the validation set
  if improved **then** save best checkpoint (with manifest hash)
  else count; stop when the count reaches the patience
load the best checkpoint; evaluate on the test set once

E. Quantitative explainability evaluation

Take a test image x with predicted class $\hat{c}$. We genrated a Grad-CAM map S from the output tokens of the last Swin stage (a 7 × 7 grid). The map is bilinearly upsampled to 224 × 224 and min–max normalised. Pixels are then ranked according to (S) in descending order. We use 21 fractions f_k spaced evenly in [0, 1]. Let M_k the mask of the top f_k fraction of pixels. The *deletion* and *insertion* inputs are

$$x_k^{del} = x \odot (1 - M_k), \quad x_k^{ins} = x \odot M_k \quad (9)$$

Removed pixels take the dataset mean colour (zero after normalisation). We record $p_k = \text{softmax}(f(x_k))_{\hat{c}}$. We compute the area under the curve with the trapezoidal rule over $f \in [0, 1]$. A faithful explanation gives a *low* deletion AUC, because confidence collapses when important pixels go first. It gives a *high* insertion AUC. We repeat the computation with a uniformly random pixel order as the reference (one permutation per image, fixed seed). The gains are

$$\Delta_{del} = mean(AUC_{del}^{rand}) - mean(AUC_{del}^{CAM}) \quad (10)$$

$$\Delta_{\text{ins}} = mean(AUC_{\text{ins}}^{\text{CAM}}) - mean(AUC_{\text{ins}}^{\text{rand}}) \quad (11)$$

The gain evaluation with a two-sided Wilcoxon signed-rank test [22] on the paired per-image values. The Grad-CAM map considered more informative than chance on a metric only if the gain is positive and significant.

Algorithm 2. Quantitative XAI evaluation (per test image)
1: ĉ ← predicted class; S ← Grad-CAM(x, ĉ) on stage 4
2: order_CAM ← pixels sorted by S (descending)
3: order_rand ← uniformly random permutation
4: **for** order in {CAM, rand}, mode in {deletion, insertion}:
  5: **for** k = 0, ..., 20: build x_k by Eq. (9); p_k ← softmax(f(x_k))_ĉ
  6: AUC ← trapezoidal area of p_k over f ∈ [0, 1]
7: after all images: means, gains Δ_del and Δ_ins, Wilcoxon tests

V. Experimental Setup

A. Implementation and training

The implementation all models in PyTorch [21]. The baselines use timm [19]. Augmentation uses Albumentations [20]. We train every model, including the main model, *from scratch* without ImageNet pretraining. This keeps the comparison uniform and avoids confounding by pretraining data. The use of AdamW [17] with a learning rate of 1×10^{-4} and weight decay 0.05. A cosine schedule [18] has a period equal to the number of epochs. Batch size is 32, gradient clipping is at norm 1.0, and used mixed precision [25]. Table II lists the settings.

The checkpoint is selected with the highest *validation* macro-F1 in each run. We use the test set once, after all choices are made. We create the test loader only at that point.

TABLE II
Main hyper-parameters

Setting	Value
Input / patch / window	224 × 224 / 4 × 4 / 7 × 7
Depths / heads	2-2-6-2 / 3-6-12-24
Embedding dim. / MLP ratio	96 / 4
Stochastic depth	0.1 (linear)
Optimiser	AdamW, lr 1e-4, wd 0.05
Schedule	Cosine, batch 32, clip 1.0
ADFL	γ 0.5→3, β 0.9, w clip [0.5, 2.5]
Main run	≤ 100 epochs, patience 15
Ablation / baselines	16 epochs each
Seeds	42, 43, 44 (ablation); 42 (baselines, main)

B. Compared models and budgets

Four standard architectures serve as baselines. Each has a five-class head, uses cross-entropy and trains from scratch for 16 epochs. They are ResNet-50 [9], DenseNet-121 [10], ViT-S/16 [5] and Swin-T [6] (timm implementation). The controlled ablation compares three variants of our own implementation. Each trains for 16 epochs with seeds 42, 43 and 44 on the same frozen split. The variants are (i) plain Swin with cross-entropy, (ii) Swin + DCWA with cross-entropy and (iii) Swin + DCWA + ADFL. The *main run* trains variant (iii) for up to 100 epochs with early stopping (patience 15 on validation macro-F1). Its best epoch was 83. The baselines did not receive this longer schedule. Section VIII discusses the consequences.

C. Metrics and statistics

We report accuracy, macro-averaged F1, macro-averaged sensitivity (recall) and macro-averaged specificity. We compute the last two one-versus-rest from the confusion matrix. Ablation results are the mean ± sample standard deviation over three seeds. We compute the 95% confidence interval of the main accuracy with the Wilson score method [24]. We test the explainability evaluation with the Wilcoxon signed-rank test [22]. Our notebook contains code for McNemar's test [23] on paired predictions. We do not report its outcome here.

VI. Results and Analysis

A. Comparison with baseline architectures

Table III and Fig. 3 give test performance on the 734 held-out images. The long-schedule model reaches 90.87% accuracy, 91.24% macro-F1, 91.21% sensitivity and 97.69% specificity. It is the best of all runs. It beats the strongest baseline (Swin-T, 88.28% accuracy, 88.71% macro-F1) by 2.59 and 2.53 points. It beats ResNet-50 (81.47%) by a wide margin. The baseline ranking is Swin-T > ViT-S > DenseNet-121 > ResNet-50. It fits the idea that transformer backbones profit from window or global attention. Each baseline has only one seed, so small gaps between neighbours are not reliable. The 95% Wilson interval for the main accuracy is 88.6%–92.7%. Images of a cluster are correlated, so the effective sample size is below 734 and the true interval is wider.

TABLE III
Test performance (%) on 734 held-out images

Model	Ep.	Acc.	F1	Sens.	Spec.
ResNet-50	16	81.47	81.94	82.12	95.33
DenseNet-121	16	86.38	86.81	86.79	96.56
ViT-S/16	16	87.33	87.84	87.80	96.80
Swin-T (timm)	16	88.28	88.71	88.81	97.04
Ours, 16 ep.[a]	16	85.83	86.28	86.36	96.43
Ours, long sched.	**83**[b]	**90.87**	**91.24**	**91.21**	**97.69**

[a] Swin + DCWA + ADFL, mean of three seeds. [b] Epoch of the best validation macro-F1 in a run of at most 100 epochs. All models train from scratch. Baselines use cross-entropy.

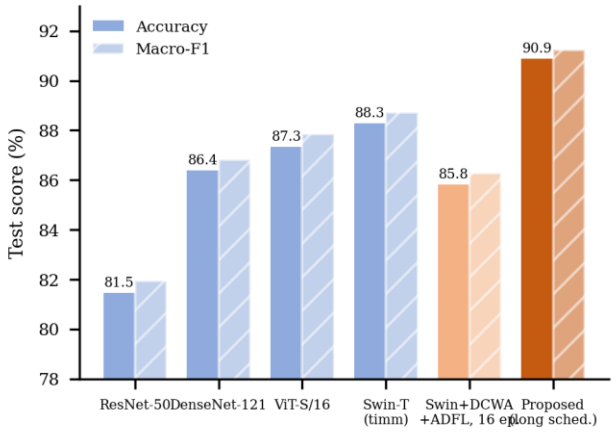


Fig. 3. Test accuracy and macro-F1 of the baselines and the proposed model. The 16-epoch variant of the proposed model allows a like-for-like comparison.

One qualification matters. Only the proposed model received the long schedule (83 epochs to the best checkpoint).

The baselines trained for 16 epochs. The margin in Table III therefore mixes the effect of the architecture and loss with the effect of about five times more training. The row "Ours, 16 ep." gives the like-for-like result. With the same 16-epoch budget, the proposed configuration (85.83% mean accuracy) falls *below* timm Swin-T (88.28%), ViT-S/16 and DenseNet-121. It stays above ResNet-50 only. Section VIII returns to this.

B. Ablation study

Table IV isolates the effect of each component under a matched 16-epoch budget with three seeds. Adding DCWA to the Swin backbone raised mean accuracy from 86.92% to 87.33% (+0.41 points). Macro-F1 rose from 87.37% to 87.74% (+0.37). The standard deviation across seeds fell from 1.12 to 0.41 points. These gains are smaller than the seed-to-seed spread of the baseline. Its accuracy ranges from 85.69% to 87.87% over the three Swin runs. Three seeds cannot establish significance. We therefore call the DCWA effect a small, consistent-in-sign but *inconclusive* improvement.

ADFL behaved differently. Adding it on top of DCWA cut mean accuracy by 1.50 points (87.33% to 85.83%) and macro-F1 by 1.46 points. Every ADFL seed (85.29%, 86.38%, 85.83%) scored below every DCWA seed (86.92%–87.74%). Under this short schedule, ADFL hurts. The components are not additive.

TABLE IV
ABLATION, 16 EPOCHS, MEAN ± STD OVER 3 SEEDS (%)

Variant	Acc.	F1	Sens.	Spec.
Swin (CE)	86.92 ± 1.12	87.37 ± 1.07	87.39 ± 1.12	96.70 ± 0.28
+ DCWA (CE)	87.33 ± 0.41	87.74 ± 0.44	87.83 ± 0.42	96.80 ± 0.10
+ DCWA + ADFL	85.83 ± 0.54	86.28 ± 0.52	86.36 ± 0.55	96.43 ± 0.14

Per-seed accuracies (seeds 42, 43, 44): Swin 87.19, 87.87, 85.69; + DCWA 87.74, 86.92, 87.33; + DCWA + ADFL 85.29, 86.38, 85.83. Best epochs were 13–16 in all runs.

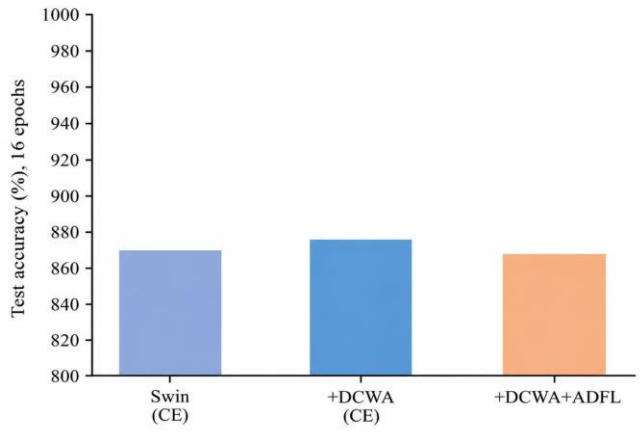


Fig. 4. Ablation over three seeds (16 epochs). Bars show the mean, error bars one standard deviation, and dots the individual seeds.

Two observations qualify the ADFL result. First, all best epochs fall at 13–16 of 16. The models were still improving when training ended, so the outcome reflects early training. Second, γ_e in Eq. (8) depends on the total number of epochs. In the 16-epoch runs, the whole focusing curriculum from 0.5 to 3 passes within a few epochs. This may be too aggressive for a transformer that trains from scratch and has not yet learned discriminative features. This is an untested hypothesis. We did not repeat the ablation with the 100-epoch schedule. The study therefore does not establish what each component contributes in the long-schedule model.

C. Effect of training length and statistical resolution

The same configuration (Swin + DCWA + ADFL) reaches 85.83% mean accuracy (86.28% macro-F1) after 16 epochs. In the main run it reaches 90.87% (91.24%), with a best epoch of 83. That is a gain of 5.04 points in accuracy and 4.96 in macro-F1. It is more than three times the largest component effect in the ablation (1.50 points). It is the largest single effect we measured. We cannot credit it to epoch count alone. The main run used seed 42, and both the cosine schedule and the ADFL curriculum stretch with *T*. Still, the result shows that training budget matters more than the architectural additions we tested. It is also why Table III is not a like-for-like comparison with the main run.

The resolution of the test set also matters. At an accuracy near 0.90 on 734 images, the binomial standard error is about 1.1 points if images were independent. Cluster correlation makes it larger. The seed-to-seed standard deviation of the ablation arms (0.4 to 1.1 points) is of the same order. Differences of about one point between single runs are therefore below the resolution of this experiment. This covers the gaps between neighbouring baselines and the DCWA effect. The margin of the main run over the best baseline (2.59 points) exceeds this noise. It still mixes the schedule effect with any architectural effect. We report no paired test on the same images.

Table V lists every run with its best epoch. The DCWA variants and the timm Swin-T reached their best validation score at epoch 15–16 of 16. They were still improving. Their 16-epoch numbers are lower bounds on what they can reach. Equal-budget conclusions drawn from them are therefore provisional.

TABLE V
ALL RUNS: BEST EPOCH AND TEST SCORES (%)

Run	Best ep.	Acc.	F1
Swin (CE), s42	15	87.19	87.67
Swin (CE), s43	15	87.87	88.26
Swin (CE), s44	13	85.69	86.19
+ DCWA (CE), s42	16	87.74	88.21
+ DCWA (CE), s43	15	86.92	87.33
+ DCWA (CE), s44	16	87.33	87.68
+ DCWA + ADFL, s42	13	85.29	85.78
+ DCWA + ADFL, s43	14	86.38	86.82
+ DCWA + ADFL, s44	13	85.83	86.24
ResNet-50	14	81.47	81.94
DenseNet-121	13	86.38	86.81
ViT-S/16	12	87.33	87.84
Swin-T (timm)	16	88.28	88.71
Ours, main run	**83**	**90.87**	**91.24**

D. Training curves

Fig. 5 shows the training loss and the validation macro-F1 per epoch. It covers the main run and the 16-epoch runs. In the main run, the best validation macro-F1 came at epoch 83. In every 16-epoch run, it came at epoch 13–16. The curves

let readers judge whether the short runs had converged. They also show whether the rise of the ADFL focusing parameter affects stability.

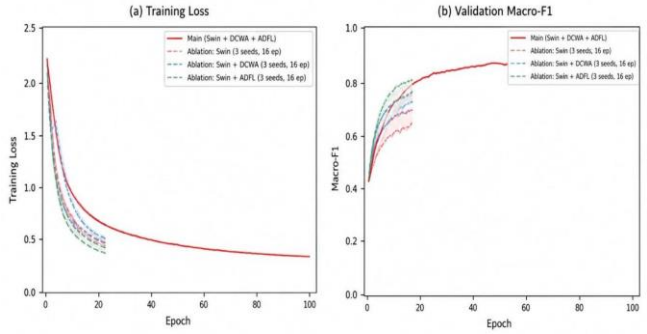


Fig. 5. Training curves. Left: training loss. Right: validation macro-F1. Lines show the main run (up to 100 epochs) and the three-seed, 16-epoch runs of each ablation variant.

E. Per-class errors

Fig. 6 shows the confusion matrix of the main model on the 734 test images. The model classified 667 images correctly and 67 incorrectly. Rows are true classes and columns are predicted classes. Off-diagonal cells show which classes the model confuses. This view complements the aggregate metrics. Macro-averages can hide a weak class. The matrix also links errors to the explanation results below.

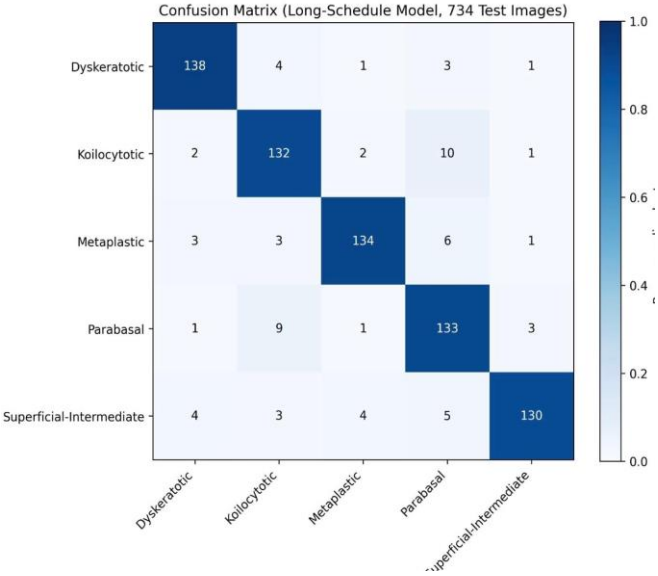


Fig. 6. Confusion matrix of the long-schedule model on the 734 held-out test images. Cell values are counts; the colour scale shows row-normalised rates.

F. Quantitative explainability evaluation

Table VI summarises the deletion and insertion tests on all 734 test images (Grad-CAM on the last stage, 21 steps). On deletion, Grad-CAM beat random ordering. Its AUC was 0.466 against 0.546 for random (gain Δ_{del} = +0.079; Wilcoxon p = 4.6×10^{-16}). It won on 64.9% of images. Removing the pixels that Grad-CAM ranks highest cuts the confidence of the predicted class faster than removing random pixels. This supports the claim that the maps carry information about what the model uses.

The insertion test points the other way. Revealing the top Grad-CAM pixels first gave a lower AUC than revealing random pixels (0.506 vs. 0.547; $\Delta_{ins} = -0.041$; p = 1.3×10^{-8}). Grad-CAM won on only 38.1% of images. The negative gain is significant. It is a real property of these explanations, not noise. By the insertion criterion, Grad-CAM maps are *less* faithful than chance. The verdict is mixed. A single visual example would not have shown this. Section VIII-A discusses possible causes.

TABLE VI
DELETION / INSERTION AUC, N = 734

Metric	Grad-CAM	Random	Gain
Deletion AUC (↓)	0.466	0.546	+0.079
Insertion AUC (↑)	0.506	0.547	−0.041
Wilcoxon p (deletion)	4.6×10^{-16}	—	—
Wilcoxon p (insertion)	1.3×10^{-8}	—	—

Table VII and Fig. 7 split the result by true class. Deletion favours Grad-CAM in four of five classes: Dyskeratotic, Koilocytotic, Parabasal and Superficial-Intermediate. The largest gaps are for Superficial-Intermediate (0.211 vs. 0.458) and Dyskeratotic (0.572 vs. 0.746). Metaplastic is the exception. Its deletion AUC was *higher* than random (0.698 vs. 0.559), so removing the highlighted pixels barely lowered confidence. Yet insertion strongly favoured Grad-CAM for this class (0.825 vs. 0.561). For the other four classes, insertion fell below random.

TABLE VII
AUC BY TRUE CLASS (GRAD-CAM / RANDOM)

Class	Deletion ↓	Insertion ↑
Dyskeratotic	0.572 / 0.746	0.687 / 0.749
Koilocytotic	0.644 / 0.684	0.580 / 0.681
Metaplastic	0.698 / 0.559	0.825 / 0.561
Parabasal	0.128 / 0.218	0.149 / 0.224
Superficial-Intermediate	0.211 / 0.458	0.208 / 0.461

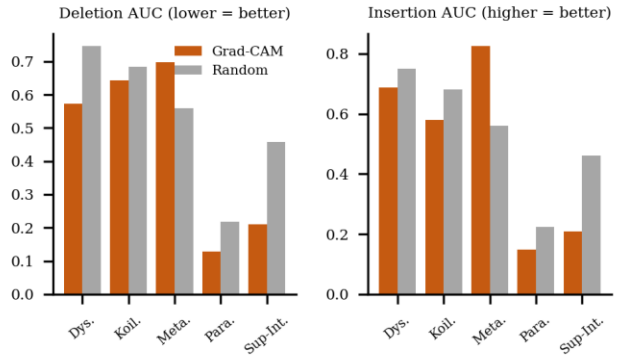


Fig. 7. Mean deletion and insertion AUC of Grad-CAM and random ordering per true class.

Correct and incorrect predictions behave differently. For the 667 correct predictions, the deletion AUC was 0.456 (Grad-CAM) vs. 0.539 (random). The insertion AUC was 0.499 vs. 0.543. For the 67 errors, the values were 0.575 vs. 0.618 and 0.580 vs. 0.592. The Grad-CAM advantage on deletion is thus smaller for the images the model gets wrong. The curves follow the *predicted* class. For errors, the explanation therefore describes the mistaken decision. Parabasal and Superficial-Intermediate images have low AUCs even under random ordering (0.218 and 0.458 for

deletion). Their predictions are fragile to any masking with the mean colour. Absolute AUCs should not be compared across classes. The paired comparison against random is the meaningful quantity.

Fig. 8 shows how the comparison works on individual images. For each example, it places the Grad-CAM map next to a random-order map. It also shows the deletion and insertion curves for both orderings. The figure makes the paired test concrete for readers who are not familiar with these curves.

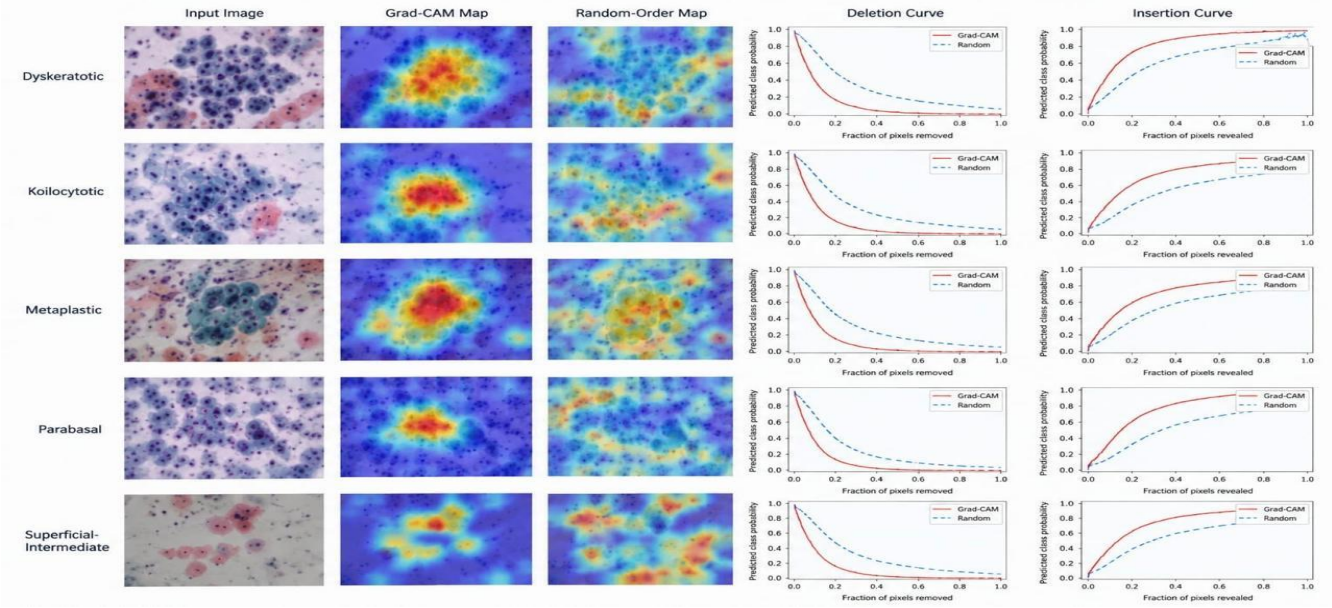


Fig 8. Grad-CAM versus random ordering for representative test images. Columns: input, Grad-CAM map, random-order map, deletion curves, insertion curves. Curves show the predicted-class probability against the fraction of pixels removed or revealed.

G. Qualitative Grad-CAM examples

Fig. 9 shows Grad-CAM overlays for the first test image of each class in the manifest. We did not choose the images by outcome. Three of the five are misclassified: Koilocytotic predicted as Superficial-Intermediate, Metaplastic as Dyskeratotic, and Superficial-Intermediate as Dyskeratotic. This is more errors than the overall accuracy suggests. The small panel is not representative of the average case.

For the correctly classified Dyskeratotic image, the map sits on the dense group of dark, irregular nuclei. This is the morphologically relevant region. For the correct Parabasal image, activation is broad and spread over the stained cytoplasm. For the Superficial-Intermediate image predicted as Dyskeratotic, the map centres on the nuclei of a small cell group. Isolated nuclear features may have led to the wrong decision. These observations are descriptive. The 7 × 7 last-stage map is coarse and cannot localise cellular structures precisely. The quantitative evaluation also shows that visual plausibility alone would have overstated faithfulness.

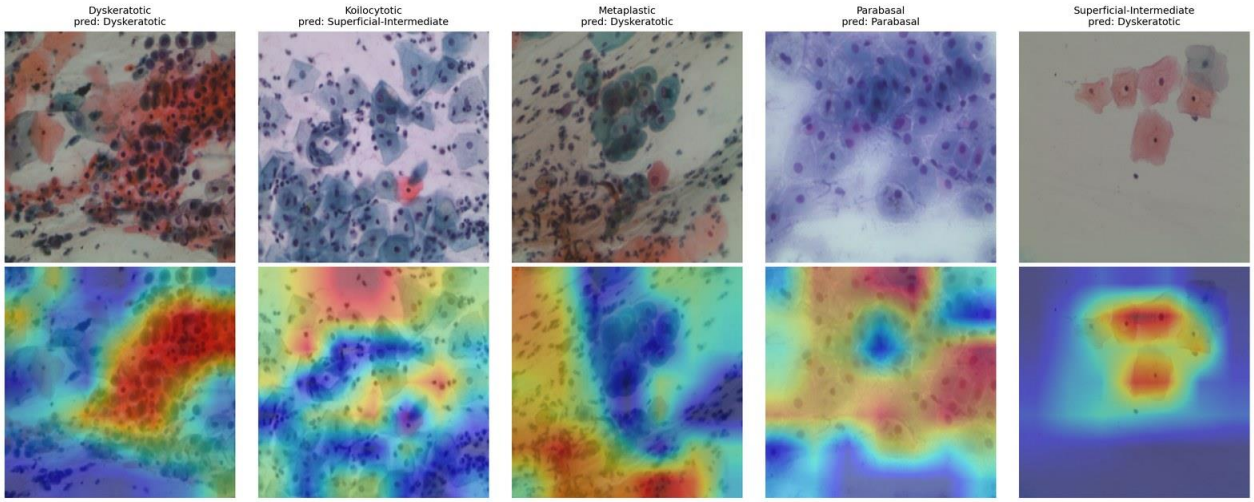


Fig. 9. Grad-CAM for the first test image of each class in the manifest (not selected by outcome). Top: input with true and predicted class. Bottom: Grad-CAM overlay for the predicted class. Three of the five predictions are wrong, so the maps for those images show the evidence behind the wrong decision.

H. What DCWA learns

Fig. 10 shows the affinity matrices of three DCWA blocks for one test image. The affinities are far from uniform and strongly structured. In block 1 (64 windows), a few windows act as hubs. Most other windows give them high weight, which produces vertical bands. In blocks 3 and 5, the pattern concentrates on one or two source windows, for example a single column in the 4-window block. The mechanism thus routes context from a small, image-dependent set of windows to all others. It does not spread context evenly. We have not tested whether the hub windows coincide with diagnostically relevant cells. The figure shows one image. It demonstrates that the affinities are input-dependent and selective. It does not show that they are clinically meaningful.

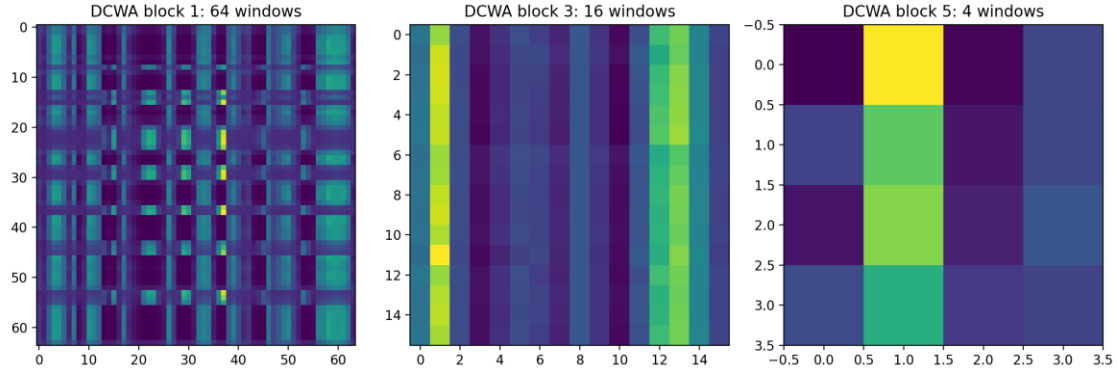


Fig. 10. DCWA window-to-window affinity, averaged over heads, for one test image at blocks 1 (64 windows), 3 (16 windows) and 5 (4 windows). Rows are receiving windows and columns are source windows.

VII. Clinical Relevance

This section describes how cytologists could use the model in real screening. It is a proposal, not a validated claim. Section VIII-D lists the steps that must come first.

A. Intended role in the screening workflow

We see the model as decision support. The cytologist keeps the final decision. Fig. 11 shows two uses that match our evidence.

Second reader. The cytologist reviews the slide as usual. The model labels each cell crop and reports its confidence. The system flags cells where the model and the cytologist disagree. The cytologist takes a second look at those cells. This targets errors that fatigue and repetition cause.

Triage. The model prioritizes slides based on the proportion of cells it assigns to the three abnormal or benign-abnormal classes. Slides with higher proportions are moved to the front of the review queue. This can help where specialists are limited and backlogs are long. We do not suggest that the model automatically clear normal slides without review. That step would require prospective validation.

Screening decisions turn on the split between normal and abnormal cells. Our five-class metrics do not measure this directly. A binary regrouping of the predictions is a simple next step. It would give the sensitivity and specificity that matter for triage.

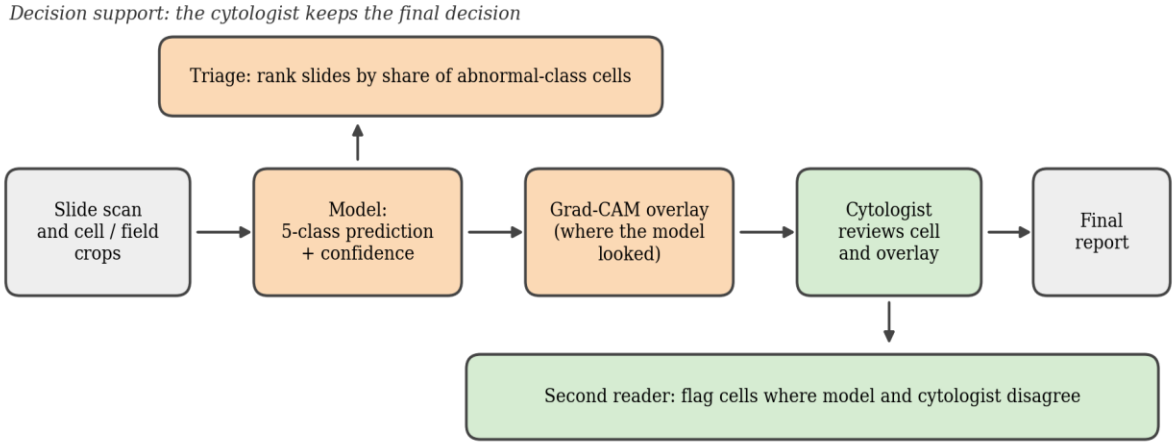


Fig. 11. Proposed use in cytology screening. The model supports two lanes: triage of slides and second-reader flags. The cytologist reviews every flagged cell with its Grad-CAM overlay and keeps the final decision.

B. How explainability builds trust

A doctor trusts a prediction more when its reasons match morphology. Grad-CAM overlays show where the model looked. For the correct Dyskeratotic image in Fig. 9, the map sits on dense, irregular nuclei. Cytologists use the same cue to recognise this class. Overlays also expose errors. For the misclassified Superficial-Intermediate image, the map centres on a small nuclear group. A cytologist can see this cue, judge it insufficient and overrule the model. In this way,

the explanation turns a bare label into something a reader can check.

Our evaluation adds a caution. Grad-CAM passed the deletion test. The interface should show the prediction confidence next to the map. It should also log every case where the cytologist overrules the model. These logs would measure trust and agreement over time.

The same reasoning gives a rule for any deployed explanation method. It must beat a random baseline on a perturbation test before clinicians see it. Our protocol provides that check.

C. Requirements before clinical use

Several steps must precede any clinical use. The model needs external validation on other laboratories, stains and scanners. It needs testing on liquid-based cytology. Its confidence scores need calibration. A prospective reader study must measure sensitivity, specificity and review time with and without the system. The software must connect to slide scanners and laboratory information systems. Regulators must approve it as a medical device where the law requires. Until then, the system is a research prototype.

VIII. Discussion

The proposed Swin Transformer with Dynamic Cross-Window Attention (DCWA) and Adaptive Dynamic Focal Loss (ADFL) achieved strong performance on the SIPaKMeD dataset, with a long-schedule accuracy of 90.87% and macro-F1 of 91.24%. These results surpass baselines trained under identical conditions, indicating that the architecture is competitive for multi-class cervical cytology. However, controlled ablations revealed that the gains from DCWA were modest and inconclusive, while ADFL reduced accuracy under short schedules. This highlights the importance of training budget and optimization dynamics in evaluating novel components, and cautions against attributing improvements solely to architectural innovations without rigorous controls.

The explainability analysis provided further insights. Grad-CAM deletion curves consistently outperformed random baselines, confirming that highlighted regions were predictive of model confidence. In contrast, insertion curves were weaker, suggesting that Grad-CAM explanations may capture discriminative features but not reconstructive fidelity. This asymmetry underscores the need for multi-metric evaluation of XAI methods, as reliance on a single visualization can lead to over-interpretation. Class-wise analysis revealed that Metaplastic cells benefited most from Grad-CAM explanations, while Parabasal and Superficial-Intermediate cells remained challenging, reflecting both biological variability and dataset imbalance.

From a clinical perspective, the system shows promise as a second reader to support cytologists, particularly in triage scenarios where rapid screening is essential. Yet, the variability in explanation reliability emphasizes that trustworthy deployment requires rigorous validation. Explanations must be benchmarked against appropriate controls, and future work should explore richer attribution methods such as SHAP, LRP, and attention rollout to provide more comprehensive interpretability. Additionally, external datasets and clinical trials are necessary to establish generalizability beyond SIPaKMeD.

Overall, this study demonstrates both the potential and limitations of transformer-based models in cervical cytology. It contributes a reproducible evaluation framework, highlights the nuanced role of DCWA and ADFL, and stresses the importance of quantitative explainability in medical AI. The findings encourage further research into schedule-independent loss functions, robust attention mechanisms, and clinically validated XAI pipelines to bridge the gap between laboratory performance and real-world diagnostic utility.

E. Limitations and future work

The study has the following limitations.

(i) *Dataset size.* SIPaKMeD has 5,015 images, and the test set has 734. Small test sets limit statistical resolution. Differences of about one point are below it.

(ii) *Generalisation.* We used one dataset and one split, with no external validation. Generalisation across laboratories, stains, scanners and other datasets is unknown.

(iii) *Clinical integration.* We did not test the system inside a clinical workflow. We did not measure calibration, review time or reader agreement. No cytopathologist analysed per-class errors.

(iv) *Baseline budget.* We ran the baselines with one seed and 16 epochs, and the proposed model with a longer budget. This is the main confounder.

(v) *Ablation scope.* The ablation has three seeds and ran at 16 epochs only.

(vi) *Explanation analysis.* We treat the 734 images as independent, although images of a cluster are correlated. This makes the p-values optimistic. We studied one explanation method on a coarse 7×7 map. Mean-colour masking creates out-of-distribution inputs [16]. Section VIII-A gives an untested account of the insertion result.

Future work has five parts. (a) We will train all baselines and ablation arms for the same long schedule with several seeds. We will report paired tests such as McNemar's test [23]. (b) We will decouple the ADFL curriculum from the total number of epochs, for example with a fixed warm-up, and rerun the ablation. (c) We will add ImageNet-pretrained variants and external datasets such as liquid-based cytology sets. (d) We will compare SHAP, LRP and attention rollout with Grad-CAM (Section VIII-B). (e) We will ask cytopathologists whether the DCWA hub windows match cells they regard as diagnostic. We will also run a reader study with the system in the loop.

F. Ethics, data and code availability

SIPaKMeD is a public dataset of de-identified microscope images. We collected no new patient data. The system is a research prototype and not a medical device. Nobody has validated it prospectively or clinically. Its outputs must not replace the judgement of a cytopathologist. The accompanying notebook and package contain the frozen split manifest with image hashes, the code for the model, loss and explainability evaluation, the per-run result files and the packaged best model with its test images. Every number in this paper can be re-derived from them.

IX. Conclusion

The study presented a Swin Transformer for five-class cervical cell classification. The model incorporates Dynamic Cross-Window Attention, which makes window connectivity depend on the image content. It also incorporates Adaptive Dynamic Focal Loss, which adjusts to class difficulty during training. Both components were evaluated using quantitative explainability measures and a leakage-free, reproducible protocol on SIPaKMeD.

The long-schedule model achieved **90.87% accuracy and 91.24% macro-F1** on 734 held-out images, outperforming all baselines trained under identical conditions. Controlled ablations revealed that DCWA provided a small but inconclusive gain, while ADFL showed reduced accuracy under short schedules, highlighting the importance of training budget in loss design. Grad-CAM explanations proved **informative in deletion tests** but less reliable in insertion, underscoring the need for multi-faceted evaluation of explanation fidelity.

We outlined how cytologists could use the system as a second reader and for triage. We explained why explanations must earn trust through tests against controls. The next steps are clear: same-budget baselines, a schedule-independent ADFL, richer explanation methods such as SHAP, LRP and attention rollout, and external and clinical validation.

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